



Department of Information Technology
Academic Year 2025 – 2026 (Even Semester)

Year, Degree, & Branch : VI-Semester-, B.Tech-IT
Course Code & Title : CCS374- & Web Application Security
Name of the Faculty member : B.Thevahi, AP/IT
Innovative Practice Description

- **Unit / Topic** : V/ Cross-Site Scripting(XSS)
- **Course Outcome** : CO5
- **Topic Learning Outcome:** TLO1&TLO8
- **Activity Chosen** : Flipped Class Room
- **Justification:**

In this activity, students explore the basic concepts before class through videos or reading materials, allowing classroom time to be used for deeper discussions. The topic Cross-Site Scripting (XSS) involves multiple concepts such as types of XSS (Stored, Reflected, DOM-based), attack flow, vulnerabilities, and prevention techniques, which can be complex for students to understand through traditional lecture methods.

Time Allotted for the Activity: 15 Minutes

❖ **Details of the Implementation:**

- Attack mechanisms, and prevention techniques, for students to study before coming to class.
- The faculty then instructed the students to take the class based on their own understanding.
- Ms Jeyadevi conducted the session on the mechanisms of attacks and explained the key concepts clearly as shown in Figure1. Subsequently, Ms. Sobika Joleen G.J. presented the types of Cross-Site Scripting as shown in Figure2.
- Before concluding the activity, students were encouraged to ask questions, and their doubts were clarified.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO6	PSO2	PSO3
CO5	3	2	3	2	3	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO5	PO6	PSO2	PSO3
CO5	3	2	3	2	3	2
Justification for correlation	Students will be able to understand and apply the basic concepts of Cross-Site Scripting (XSS), including its types, attack mechanisms, and prevention techniques used in web security.	Students will be able to analyze security vulnerabilities in web applications and identify how XSS attacks exploit user inputs, leading to data theft and unauthorized access.	Through the flipped classroom activity, students actively explore real-time scenarios of XSS attacks and prevention methods.	Students will understand the importance of secure coding practices and ethical responsibility in preventing XSS attacks.	Students will be able to clearly present and explain XSS concepts, attack flows, and preventive measures during peer teaching, improving their communication and presentation skills.	The activity encourages self-learning, as students prepare in advance and stay updated with modern web security threats and defense techniques, supporting lifelong learning.

- Images / Screenshot of the practice:



Figure1: Flipped Class Room Presented by S.Jeyadevi **Figure2: Presented by Sobika Joleen G.J**

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- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students felt that learning the content before class and discussing it during class made the topic easier and more engaging.
- This activity enhanced student participation and improved overall understanding of web security concepts.

- ❖ ***Benefit of the practice:***

- The flipped classroom model allows students to learn XSS concepts at their own pace before class.
- Class time is effectively used for discussion, clarification, and practical understanding.
- It promotes active learning, critical thinking, and problem-solving skills.

- ❖ ***Challenges faced in implementation:***

- Some students did not prepare in advance, which affected their participation in classroom discussions
- A few students needed extra guidance to create clear and meaningful mind maps.
- Understanding real-time attack scenarios and coding examples of XSS was challenging for beginners.
- Some students needed repeated explanations to clearly understand prevention techniques like input validation and output encoding.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/20252026/KP_CS3391_Flipped%20Class%20Room.pdf
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/SS_IT3501_Flipped%20Class_U2.pdf

Signature of Faculty Member

HOD